Research on Practical Privacy-Preserving Machine Learning

Privacy-enhancing technologies let us extract value from sensitive data without exposing individuals. We present practical advances in federated learning and noise injection under secure aggregation — and how combining them yields better privacy–utility trade-offs. With tightening regulation and growing stakes around personal data, demonstrable privacy protection is no longer optional.

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Are you managing the privacy risk of your AI?

The Health records, financial data, and personal communications hold enormous value for research and innovation — but organisations that use them face growing regulatory requirements and reputational risk. The EU's AI Act, EHDS, and GDPR increasingly demand demonstrable evidence that individuals are protected, not just policy commitments. Privacy-enhancing technologies (PETs) provide the technical means to meet these demands.

Differential Privacy and

Noise injection to reduce traceability of training data.

Problem: An adversary can use Membership Inference Attacks (MIA) determine whether a specific individual's data was used to train a model — a privacy violation under GDPR and EHDS.

The standard defence: Differentially private stochastic gradient (DP-SGD) descent adds calibrated noise during training, providing formal (ε,δ)-guarantees. It has challenges:

- Steep accuracy decline — especially for CNNs, where architectural modifications (GroupNorm) alone cost.
- Computational cost — per-sample gradient clipping is expensive; hyperparametes change:larger batch size

Our approach: Instead of blanket noise, Forgotten-by-Design (FbD) method identifies and downweights the most vulnerable training samples using per-instance vulnerability scores. Targeted protection where it matters. Though no formal (ε,δ)-guarantees — empirical privacy via multi-attack evaluation

Why is Forgotten-by-Design practical?

- No modifications to model architecture required
- Much better accuracy–privacy trade-off than DP-SGD
- Regulators ask for demonstrable privacy protection

Federated Learning with Heterogeneous Data

Keep data at the source — bring the computation to the data.

Problem: Data heterogeneity across sites (hospitals, regions, devices) degrades federated models. Existing approaches either ignore client differences (FedAvg), require costly cluster discovery (IFCA, DAC), or maintain per-client models (Ditto).

Our approach: Each client condition their training of the global model on training data statistics computed locally at each client. Zero extra communication.

Experiments E1–E4:

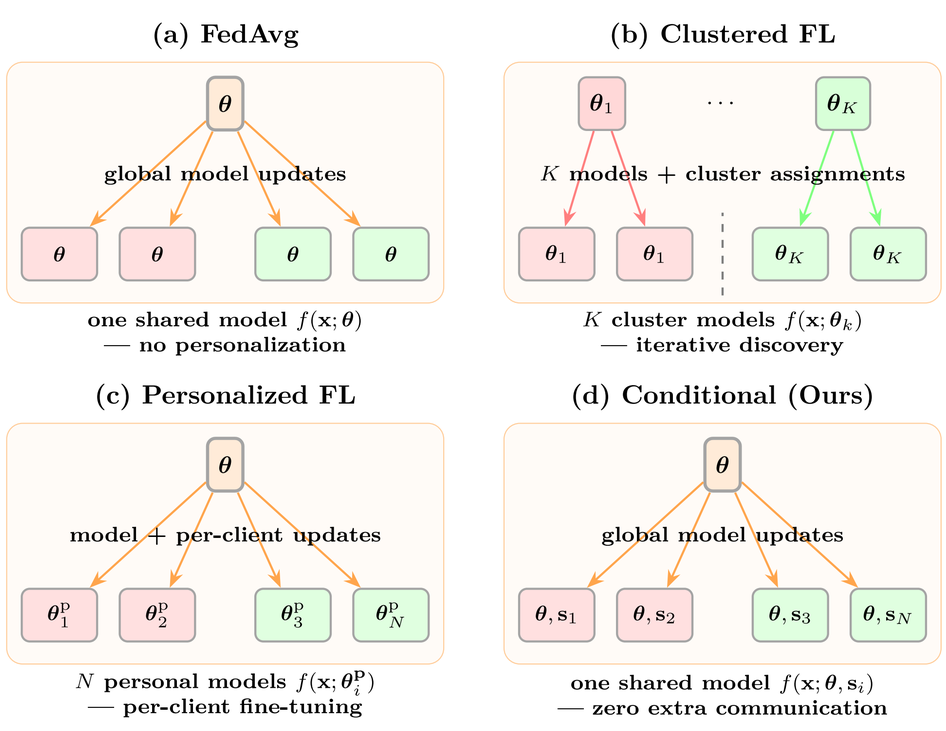
- E1 Label shift — clients see different subsets of classes
- E2 Covariate shift — clients see different input distributions (subsampling, rotation)
- E3 Concept shift — same labels map to different visual concepts (semantic, permutation)
- E4 Combined heterogeneity — multiple shifts simultaneously

Evaluated across 97 configurations: 4 heterogeneity types × 4 datasets × 7 baselines × 5 sparsity levels.

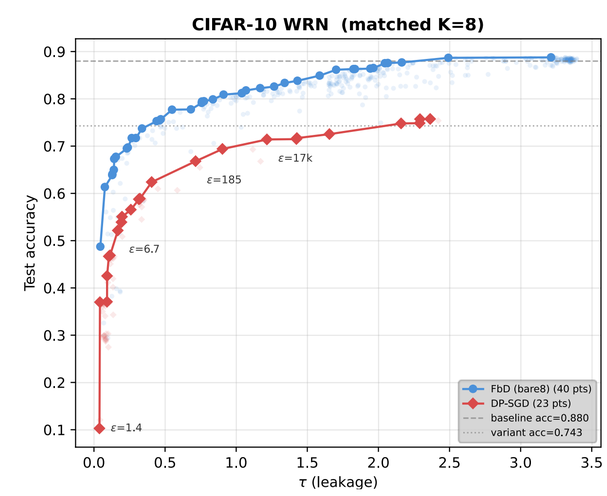
Exp.	Config	FedAvg	Oracle	Cond	Best Other
E1	CIFAR-10 K=5	.325	.927	.929	.906 (IFCA)
E2a	CIFAR-10 K=3	.943	.979	.981	.972 (DAC)
E2b	FMNIST rot. K=4	.789	.900	.895	.868 (DAC)
E3a	FMNIST sem. Rich	.671	.950	.950	.943 (IFCA)
E3b	CIFAR-10 perm. K=2	.408	.795	.808	.773 (DAC)
E4a	MNIST+FM Rich	.939	.957	.955	.953 (IFCA)
E4b	CIFAR-10 C=3	.743	.914	.935	.880 (Ditto)

Why is client conditional FL practical?

- No additional communication or disclosure beyond FedAvg
- No cluster discovery, no peer identification
- Only adds <1% model parameters
- Fully compatible with secure aggregation and DP-SGD



Left figure is comparing our Client Conditional method with the conventional Persolized and Clustered methods for heterogeneous federated learning. Illustrated for N clients partioned over K (unknown) clusters. Our method doesn't require peer identification or cluster discovery, and adds no communication overhead or private data disclosure compared to the vanilla Federated Averaging. The table shows that it ourperforms standard heterogeneous federated learning benchmark methods.



Forgotten by Design (FbD) instance specific noise-injection outperforms conventional Differential Privacy (DP-SGD). Figure shows utility-privacy trade-off, that is, accuracy versus traceability τ.

Other Privacy Enhancing Technologies at RISE

Fully homomorphic encryption enables end-to-end computation on encrypted data without decryption.

- Not practical for training – encrypted inference for but the simplest neural networks remain challenging.
- Our co-design of Transformer attention eliminates the most expensive operations achieving 3-6 times faster inference.

Synthetic data mimics statistical patterns in source data

- Useful for testing pipelines and interoperability across jurisdictions, but high fidelity for ML remains difficult.
- Generation from sensitive sources does not guarantee privacy — one must evidence it, e.g. using LeakPro.

Combining PETs. No single PET covers all attack surfaces.

- FHE based protocols for federated secure aggregation where partial results are combined in ciphertext are deployable today
- Secure aggregation combined with global DP achieves privacy–utility comparable to centralised processing.
- Our client-conditional FL fits naturally: statistics stay local, no peer identification needed

HOW WE CAN HELP

PET advisory. Matching privacy technologies to your data flows, threat models, and regulatory requirements (GDPR, EHDS, AI Act).

Privacy testing. Auditing models and synthetic datasets for information leakage – prepare evidence for regulatory reporting (DPIA).

Explore opportunities through the **TEFs** (Test & Experiment Facilities) and **AI Factory MIMER.**